

Two-Repository Market System - Status and Pre-Research Review

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NO MARKET ANALYSIS, SIMULATION, BACKTEST, DATA ACQUISITION, OR RESEARCH EXECUTION IS AUTHORIZED BY THIS REPORT.

Executive summary

The project currently spans two separate repositories. `version2.0` is the product dashboard and current-display system with exploratory tools and candidate backtests. `custom_terminal` is the authoritative governance, provenance, data-trust, preregistration, approval, evidence and lifecycle system. This report reconciles them without merging or copying code. It does not approve either repository's market calculations.

Together they are intended to become a personal Indian-market research system that can explain what it knows, reproduce how it knew it, test ideas honestly, and keep investment decisions separate from unverified calculations. Its eventual purpose is decision support for trading, investing, asset allocation, portfolio balance, and downside risk. It is not yet an automated trading system, and it does not currently have an approved investment edge or production score.

The development order deliberately puts trust before strategy testing. A statistically attractive result is not useful if the historical population excludes failed companies, if ticker identities are wrong, if corporate actions are missing, or if today's transaction costs are applied to old trades. The project therefore stopped momentum research when the local data was shown to be survivor-selected and incomplete.

What is working now is the research machinery: versioned contracts, immutable evidence, point-in-time concepts, deterministic tests, one-use approvals, a governed execution gateway, and an independently audited three-row synthetic canary. The canary proved infrastructure behavior only. It did not test an investment idea.

The main blocker is data trust. The local panel contains useful price history, but it is not a complete historical market population. Only three of twelve dated security-snapshot and bhavcopy pairs qualified. Stable identity, inactive securities, economic terminal outcomes, authoritative corporate actions, NIFTY PRI/TRI history, complete dated statutory costs, and access/retention rights remain incomplete.

The proposed direction is cautious: review this report first, define one bounded free-source capability question, qualify access and retention, run a small data pilot only if permitted, and only later preregister a descriptive or falsification study. A separate exact run approval would still be required.

Current overall status

Area	Status	Meaning
Architecture	ESTABLISHED	A modular local research architecture and dependency direction are defined.
Research governance	PROVEN FOR SYNTHETIC CANARY	One bounded non-market canary completed and was independently audited.
Historical equity data	NOT TRUSTED	The current panel is survivor-selected and cannot support complete-market claims.
Historical population	FAIL	Only 3 of 12 locked security/bhavcopy pairs qualified.
Market research execution	BLOCKED	This PDF is pending review; data and run-specific gates also remain closed.
Production score or recommendation	NOT AVAILABLE	No edge has passed the required validation and evidence chain.
version2.0 execution outputs	UNGOVERNED / NONCANONICAL	Useful product and exploratory work exists, but it did not run through <code>custom_terminal</code> governance.

Two-repository architecture and authority

The repositories are complementary but not equivalent. `version2.0` is a product-facing dashboard with current monitoring, data pages, exploratory backtests, live-signal computation and reusable presentation/provider code. `custom_terminal` is the only canonical authority for data provenance, point-in-time trust, historical-universe qualification, research-family registration, preregistration, input hashes, split access, exact one-use approval, immutable manifests, canonical import, evidence lifecycle and promotion.

No code or data was merged or copied during R.7. Selective integration may be proposed later, one bounded component at a time, only after owner review.

Responsibility	Current implementation	Authority	Trust status	R.7 decision
UI and report presentation	Mainly <code>version2.0</code>	<code>custom_terminal</code> published read models	Display/exploratory	Selectively retain
Current monitoring	Both repositories	<code>custom_terminal</code> scope controls	Current-only	Reconcile later
Data acquisition	Provider code in both	<code>custom_terminal</code> immutable contracts	Mixed/unqualified	Adapt selectively
Provenance and revisions	<code>custom_terminal</code>	<code>custom_terminal</code>	Canonical contract	Retain
Historical universe and identity	<code>custom_terminal</code> contracts; <code>version2.0</code> current NIFTY 500	<code>custom_terminal</code>	Current capability FAIL	Prohibit substitution
Research and backtesting	Both repositories	<code>custom_terminal</code> governed gateway	<code>version2.0</code> noncanonical	Re-evaluate only after preregistration
Approval, evidence, lifecycle	<code>custom_terminal</code>	<code>custom_terminal</code>	Canonical	Retain exclusively
Live signals and recommendations	<code>version2.0</code>	None approved	Blocked	Disable pending review
Broker and trading	<code>version2.0</code> classes	None approved	Live-action code exists	Isolate or retire

Product capability classification

These labels describe what the reviewed code can currently be trusted to do. They are not judgments about visual quality or future usefulness.

Product capability	Classification	Meaning
Streamlit dashboard and report pages	DISPLAY_ONLY	Present information; displayed claims inherit their source limitations.
Market-regime and current-monitoring views	CURRENT_STATE_DIAGNOSTIC	Current context only; not historical evidence or allocation authority.
Lab controls and interactive studies	EXPLORATORY_ANALYSIS	Useful for hypothesis development; outputs are not canonical evidence.
Validation, recommendation backtest and research scripts	UNGOVERNED_BACKTEST	Did not use the custom_terminal report, preregistration, input or approval gates.
Read-only options, macro, filing and provider presentation	CANDIDATE_FOR_GOVERNED_INTEGRATION	Potentially reusable after contract, timing, provenance and rights review.
ZerodhaBroker.place_order	LIVE_ACTION_CAPABLE	The code can submit an order if instantiated with a capable token.
Recommendations, live scans, deployment language and broker actions	BLOCKED_PENDING_REVIEW	Must not become product decisions or trading actions.

All version2.0 analysis and backtest outputs are classified UNGOVERNED_NONCANONICAL. This preserves their history without treating them as validated by the newer system.

version2.0 positive findings and reusable value

- One shared recommendation-display flag hides strategy recommendation tables.
- Reviewed results do not show a meaningful benchmark-relative edge; negative and rejected strategies remain visible instead of being erased.
- Options display queries open the local SQLite database read-only, bound expensive scans, and do not fabricate implied volatility when the source lacks it.
- Several pages disclose incomplete coverage, estimated values, assumed timing and current-snapshot limitations.
- Causal shifts exist in key feature paths, the Status and Lab responsibilities were separated, and reusable UI/provider abstractions exist.
- Current monitoring, options-query presentation, macro and filing views, provider adapters and report design are candidates for selective future integration.

These findings support reuse, not wholesale adoption. Product presentation can be good while the underlying historical research claim remains unqualified.

Material risks and contradictions

- The current NIFTY 500 list is not a historical universe. Inactive, suspended, merged and terminated securities are not reconstructible from it.
- Stable identity, terminal economics and corporate-action continuity are not qualified.
- Direct validation, backtest, refresh and scan controls sit outside the governed gateway. A hidden live-signal table may still be computed, and the hiding flag is mutable.
- Yahoo, Kite and CSV histories are not historical-qualified inputs. CSV provenance, adjustment status, corrections and revision vintages are incomplete.
- The approximately 45 GB F&O database is unqualified for provenance, coverage, correction history, retention rights and research suitability.
- Transcript, ownership, insider, ratings, AUM and index-reconstitution features need causal publication and revision timing before historical use.
- Claims such as "passed validation", "point-in-time", "historical universe" and "heldout" conflict with the canonical trust record unless re-proved under governance. A recent period is not necessarily an untouched holdout.
- version2.0 persists Kite credentials/session material, while custom_terminal deliberately uses memory-only, read-only controls.
- Broker substitution is disabled in the reviewed engine, but live order-capable code still exists and must remain isolated.

What are we trying to build?

The intended system is a trustworthy Indian-market research foundation. It should collect point-in-time evidence, preserve revisions, calculate features deterministically, define future outcomes before looking at them, validate hypotheses out of sample, and publish evidence that can be independently reconstructed.

The core chain is:

Data sources -> immutable raw evidence -> typed normalized data -> point-in-time features -> declared outcomes
-> governed experiments -> validation -> scores and confidence -> decision policy -> portfolio and risk support.

Three kinds of evidence must remain separate:

- Prediction evidence asks whether a feature ranks or forecasts later outcomes.
- Economic evidence asks whether the relationship survives execution assumptions, costs, turnover, capacity, and adverse excursions.
- Portfolio evidence asks what happens when positions interact through concentration, exposure, drawdown, cash, and constraints.

A research idea should move through exploration, validation, one-time test, independent replication, and only then production consideration. A result can be rejected at any stage. A 0-100 score, if eventually built, must be a calibrated representation of validated out-of-sample evidence. Confidence must be separate from the raw score, and the final decision must also consider regime, holdings, horizon, risk, and mandate.

The system is not yet able to claim complete-market historical results, survivorship-safe portfolio performance, delisting-aware returns, verified total-return comparisons, production-grade momentum, actionable asset scores, or live portfolio recommendations. It implements no production research approval from this PDF and no automated trade execution.

What has been completed?

Architecture review and refinement

Purpose: replace a page-by-page Streamlit scaffold and a separate script-heavy research laboratory with one modular system. Result: a local modular monolith was selected, using typed contracts, Parquet for analytical facts, DuckDB for queries, SQLite for workflow state, immutable artifacts, and a thin Streamlit layer. Significance: shared engines should serve multiple pages rather than duplicate calculations. Evidence type: architecture and infrastructure only.

Momentum Vertical Slice A

Purpose: migrate the existing 12-1 momentum experiment through reproducible data, feature, outcome, experiment, manifest, cost, and walk-forward contracts. Result: deterministic infrastructure, a golden compatibility fixture, explicit execution assumptions, manifests, and validation layers were created. The result remained non-actionable. Significance: the slice exposed trust problems instead of manufacturing a favorable edge. Evidence type: compatibility and research-process evidence; not production investment evidence.

Dataset trust and survivorship audit - Slice A.5

Purpose: determine whether the local historical panel represented the historical market. Result: NOT TRUSTED for cross-sectional equity-edge research. The 294 mapped equities behave like present-day survivors with attached back history. A separate historical reference contains 1,699 symbols absent from the panel. Significance: causal calculations inside the panel do not make the population complete. Evidence type: data-quality evidence.

Provider-neutral ingestion - Slice A.6

Purpose: prepare replaceable ingestion, typed schemas, immutable raw manifests, identity resolution, reconciliation, and dataset acceptance. Result: an end-to-end local dry run produced 1,166,839 price rows but failed the required trust capabilities. Significance: future sources can be judged by the same contract rather than trusted by reputation. Evidence type: infrastructure and data-quality evidence.

Official public-source qualification - Slices A.7 to A.10

Purpose: determine whether legally accessible official public sources can reconstruct population, identity, lifecycle, benchmarks, and costs. Result: the 12-date sample retrieved every bhavcopy but only three dated security snapshots. Direct population reconstruction failed at 3/12; event reconstruction also failed because no complete event ledger was proved. Automated NSE collection was classified as prohibited by the reviewed terms, and retention/derived-use questions require review. No substantive written NSE response was available. Significance: a public file existing on a webpage is not proof of permission, continuity, completeness, or research suitability. Evidence type: bounded source-access and sample-qualification evidence.

Security identity, corporate actions, terminal outcomes, benchmarks, and costs

Purpose: prevent ticker stitching, invented delisting values, destructive price adjustment, PRI/TRI confusion, and current-cost backfilling. Result: typed models and fail-closed checks exist, but authoritative historical evidence is incomplete. All 294 local equity identities remain unresolved. Three sampled delistings have no proven economic consideration. One symbol transition qualified; split, bonus, dividend, rights, demerger, and complete terminal quotas did not. The local NIFTY file is PRI-like only, and the historical statutory-cost schedule is incomplete. Significance: the system now says UNKNOWN or FAIL rather than guessing. Evidence type: contract and gap evidence.

Current-market Kite path - Slices A.11 and A.12

Purpose: provide a separate, read-only, in-memory current-market health path. Result: daily login, current inventory, bounded quotes, sanitization, explicit endpoint allowlist, and current-only coverage health were implemented. Significance: this helps present-time monitoring but cannot reconstruct historical populations or validate research. Evidence type: current operational infrastructure only.

Research evidence reconciliation - R.1 and R.2

Purpose: reconcile the older laboratory with the newer trust system without rewriting history. Result: 32 legacy rows were preserved: 26 rejected and 6 accepted under the old vocabulary, across 13 reviewed families. None has a verified run manifest; every row is production-ineligible. The one legacy validation-confirmed momentum row remains exploratory because it lacks the new preregistration, population, identity, terminal, cost, and test/replication chain. Significance: preservation is not retroactive validation. Evidence type: quarantined legacy evidence.

Future governance contracts - R.3 and R.4

Purpose: close canonical research execution behind complete preregistration, immutable inputs, split access, manifests, approval, and import validation. Result: the governance catalog is hash-linked; every attempt must produce an atomic bundle and root manifest; direct laboratory scripts are marked noncanonical. Exact one-use approval is mandatory, while wall-time and memory declarations remain unenforced. Significance: future evidence has a controlled lineage and cannot become canonical merely because a script produced a result. Evidence type: governance infrastructure.

Synthetic canary - R.5 and R.6

Purpose: prove the governance mechanism on three synthetic rows without market data. Result: one canary was proposed, separately approved, executed once, consumed its approval, produced three declared artifacts, created one canonical record, and passed an independent byte-level audit with zero mismatches. Its lifecycle is `INFRASTRUCTURE_CANARY_COMPLETED` and promotion eligibility is permanently false. Significance: the governance machine works for a bounded non-market case. Evidence type: infrastructure evidence only.

What do we currently have?

Application and package architecture

- A Streamlit product scaffold organized by user job.
- A provider-neutral market-intelligence package with foundation, research, simulation, evidence, portfolio, and application boundaries.
- A separate legacy Data test laboratory whose direct outputs are explicitly noncanonical.
- Versioned temporal, identity, feature, outcome, universe, cost, benchmark, experiment, manifest, approval, and governance contracts.

Data and fixtures

- A local 430-file price directory, including 294 currently mapped equities, benchmarks, and unresolved instruments.
- A normalized dry-run panel with 1,166,839 price rows.
- Twelve official bhavcopies in the locked population sample, but only three corresponding historical security snapshots.
- Small deterministic fixtures for point-in-time, identity, lifecycle, cost, benchmark, governance, and canary tests.
- A three-row synthetic canary fixture that contains no market observations.

The local historical price panel must be described as a point-in-time liquidity rule over the observed historically traded panel. It must not be described as the complete historical NSE listed population.

Governance and evidence

- A hash-linked governance event catalog.
- One independently audited, nonpromotable canonical canary record.
- A tracked sanitized canary anchor containing exact hashes.
- A preserved neutral legacy ledger and separate legacy evidence catalog.
- An entry-point inventory with zero known unsafe canonical bypasses.
- A permanent pre-research PDF policy introduced by this milestone.

Current verification baseline

- Main project suite: 230 passed, with two known deprecated-runner warnings.
- Separate Data-test suite: 289 passed, with existing third-party SWIG and noncanonical-log warnings.
- Focused R.7 gate and PDF suite: 17 passed.
- Combined R.3-R.7 governance suite: 94 passed.

What remains unresolved?

Research blockers

Blocker	Current finding	Why it matters
Historical listed population	FAIL	Missing inactive and terminated listings can create survivorship bias.
Historically traded population	UNKNOWN	Twelve sample files do not prove continuous coverage or retained revisions.
Stable security identity	FAIL	Ticker strings cannot safely join names through symbol, ISIN, merger, or listing changes.
Suspension and delisting history	UNKNOWN / FAIL	Missing states can distort eligibility, fills, marks, and returns.
Economic terminal outcomes	FAIL	Last quoted price is not delisting or merger consideration.
Corporate actions	UNKNOWN	Raw discontinuities do not prove action type, ratio, or total return.
Benchmark PRI/TRI	PARTIAL / UNKNOWN	PRI cannot be silently presented as dividend-inclusive TRI.
Historical statutory costs	INCOMPLETE	Current rates cannot be carried backward into historical trades.
Access, retention, licensing	REQUIRES REVIEW	Technical accessibility is not permission to automate or retain an archive.

Important hardening, but not the primary data blocker

- The filesystem owner can rewrite local files. Hash chains detect changes only relative to separately preserved anchors.
- Ignored raw canary evidence lacks a documented backup, retention, restore-test, and access policy.
- The declared five-second wall time and 64 MiB memory limit were not process-enforced during the canary.
- The canonical evidence catalog has one exact anchored record but no record-to-record hash chain.

These items matter before serious governed research, but fixing them would not make incomplete historical data trustworthy.

Free-source constraint and opportunities

Paid market-data providers are unavailable for this project. Free-source work must therefore separate six different questions: Is the object public? Is access lawful? Is automation permitted? Is local retention permitted? Is the history complete? Is it point-in-time correct and fit for the intended claim?

Source	Potential role	Current caution
NSE	Bhavcopies, security files, corporate filings, listing and delisting notices	Pre-February-2024 security snapshots and retention rights remain unresolved; systematic collection was not authorized.
BSE	Independent listing codes, security references, corporate actions, lifecycle evidence	BSE listings must remain distinct from NSE listings; no complete historical reconciliation is qualified.
SEBI	Regulations, orders, delisting and scheme evidence	Strong event/legal evidence, but not a complete daily exchange population.
AMFI	Mutual-fund NAVs and fund-industry information	Relevant only to future mutual-fund questions; not equity identity or survivorship evidence.
RBI	Rates, liquidity, currency, credit, and macro publications	Potential macro context; each series still needs release-vintage and revision treatment.
Government publications	Taxes, stamp duty, GST/service-tax history, official calendars	Useful for date-effective costs; documents must be assembled with supersession links.
Issuer filings	Action terms, merger/demerger schemes, consideration, dates	Case-specific supplements; conflicts cannot silently override exchange evidence.

No source is qualified merely because a webpage, download button, PDF, or current file exists. A useful future source must pass access, immutable provenance, identity, population, correction, and intended-use checks.

version2.0 execution-path inventory

Every path below can initiate a calculation, refresh, report, recommendation-like output, simulated order or live-capable order. None is canonical research.

Path or surface	Action	Classification
main.py, core/runner.py	Configured backtests, comparisons and HTML reports	UNGOVERNED_NONCANONICAL
Status page	Strategy validation, live scan, recommendation backtest and snapshot writes	UNGOVERNED_NONCANONICAL
Lab page	Interactive strategy backtest	UNGOVERNED_NONCANONICAL
Legacy directly runnable backtest page	Validation, scan, recommendation and backtest controls	UNGOVERNED_NONCANONICAL
EOD refresh widget	Equity/F&O/OI/index refresh, then regime and live-signal snapshots	UNGOVERNED_NONCANONICAL
Options page	Options database refresh and report queries	UNGOVERNED_NONCANONICAL
market_gate/research/ *.py	Attribution, blends, diffs, cost/exit sweeps, IC, reruns, neutralization and diagnostics	UNGOVERNED_NONCANONICAL
Scanner, analysis and macro CLIs	Scanner runs, Yahoo/Claude analysis and diagnostics	UNGOVERNED_NONCANONICAL
Fetch/build/clean/audit/study scripts	Data mutation, features, options universe and studies	UNGOVERNED_NONCANONICAL
SimulatedBroker	Simulated orders	UNGOVERNED_NONCANONICAL
ZerodhaBroker.place_order	Live order submission and history lookup	LIVE_ACTION_CAPABLE

The exact enumerated inventory is stored in specs/cross_repository_reconciliation_v1.json. The reviewed app currently instantiates the simulated broker, not the live broker. That reduces present wiring risk but does not make the live code safe or approved.

Selective integration strategy

The recommended strategy is selective integration, not a rewrite and not a repository merge. Candidate components are Streamlit design, read-only current-data displays, current monitoring, bounded options queries, macro and filing presentation, strategy/provider interfaces and report presentation.

Before any component enters the authoritative path it needs: a named owner; an explicit current-versus-historical scope; data provenance and availability contracts; no direct market execution; deterministic tests; and a published-read-model boundary. Existing product pages may remain as exploratory displays while this review is pending.

The following responsibilities remain exclusively in `custom_terminal`: research-family registration, preregistration, exact input hashes, the pre-research PDF gate, split access, one-use run approval, immutable manifests, canonical evidence import, lifecycle state and promotion decisions.

Retain, redesign and retire boundaries

Disposition	Components
Retain as candidates	UI design, report presentation, read-only options query patterns, current monitoring and provider interfaces
Redesign before use	timing/provenance for filings and macro, data refresh, current regime language, strategy interfaces and report labels
Quarantine as legacy	all version2.0 backtest, validation, live-scan and research results
Isolate or retire	credential/token persistence, live broker path, direct recommendation and deployment controls
Keep authoritative	all <code>custom_terminal</code> trust, governance, approval, evidence and lifecycle contracts

What should happen next?

Staged program

1. The owner reads this amended two-repository PDF and records corrections or explicit approval.
2. Resolve the owner questions on repository roles, recommendations, broker code, data rights and integration priority.
3. Freeze version2.0 market-analysis, backtest, live-scan and recommendation outputs as UNGOVERNED_NONCANONICAL evidence.
4. Define one bounded component-integration proposal with no market execution.
5. Map its inputs to custom_terminal provenance, availability and current-versus-historical contracts.
6. Remove or isolate direct execution, persistent credentials and action-oriented outputs from the proposed boundary.
7. Regenerate and re-review this PDF if any included research-state byte or reviewed repository binding changes.
8. Only after report approval, preregister one bounded descriptive or falsification study if its data capabilities pass.
9. Bind immutable inputs and obtain a separate exact one-use run approval.
10. Only then permit the governed gateway to conduct that exact approved analysis.

Go criteria

- Owner explicitly approves the exact current report and covered scope after PDF generation.
- The proposed source has documented access and retention conditions compatible with personal non-commercial research.
- The pilot question is answerable without pretending observed traded rows are the complete listed market.
- Required identities, event timing, benchmark class, cost intervals, and missing-data treatment are declared.
- Research-relevant state fingerprint still matches this report.
- A complete preregistration and separate run-specific approval exist.

Stop criteria

- CAPTCHA, authentication bypass, prohibited automation, persistent rate limiting, or unclear retention rights.
- Missing required official objects, unrecognized schema, mutable content without preserved vintages, or unresolved identity keys.
- A proposed claim requires complete historical population, delisting economics, or TRI/cost evidence that is not available.
- Any request to choose parameters, periods, or hypotheses because preliminary outcomes look favorable.
- Any attempt to treat PDF approval as run authorization.

Proposed future analysis boundary

This report covers only BOUNDED_FREE_SOURCE_CAPABILITY_PLANNING. After explicit review, that scope would permit planning and preregistration preparation, not analysis or acquisition by itself.

Questions that might eventually be supportable, subject to source-specific gates, include:

- Descriptive properties of securities actually observed trading on a session.
- Liquidity and exchange-turnover distributions within a clearly stated observed-traded population.
- Coverage, missingness, schema-transition, and identifier-quality studies.
- Signal behavior conditional on continuous observed history, clearly censored and nonpromotable.
- Infrastructure falsification tests designed to expose leakage, identity, cost, or universe failures.
- Macro descriptive studies using official release-vintage data, if revisions and publication timing are preserved.

Claims that remain prohibited include:

- Complete-market cross-sectional alpha.
- Survivorship-safe historical portfolio performance.
- Delisting-aware or merger-aware total returns.
- Production-grade 12-1 momentum.
- Actionable 0-100 scores or edge lifecycle promotion.
- Live portfolio allocation, accumulation, reduction, or avoidance recommendations.
- Automated or manual trading actions based on research output.

R.7 selects no investment hypothesis, parameters, period, security, or favorable-looking strategy.

Decisions requested from the owner

Please review and answer each question explicitly:

1. Is the project objective correct: a trustworthy research and capital-allocation support system rather than a fast signal generator?
2. Is the development order acceptable: governance and data qualification before strategy testing and scoring?
3. Is the no-paid-data constraint represented accurately and expected to remain in force?
4. Should any milestone or priority be changed before more data work?
5. Should optional governance hardening - raw-evidence backup, enforced process limits, or canonical-catalog chaining - happen before source planning?
6. Do you approve proceeding only to a bounded free-source capability-planning phase under scope `BOUNDED_FREE_SOURCE_CAPABILITY_PLANNING`?
7. What corrections or clarifications must be incorporated into a revised PDF before approval?
8. Should version2.0 remain a separate product repository or become a UI package in a later governed change?
9. Which version2.0 pages are the highest-value candidates for selective integration?
10. Should recommendation and live-scan computation be disabled, rather than merely hidden, until governed evidence exists?
11. Should live broker code be removed, quarantined, or retained in a separately approved repository?
12. May the local F&O database be retained and audited, and what evidence supports those rights?
13. Which current-data providers may be used for display-only diagnostics?
14. Should legacy "passed validation" labels be renamed `UNGOVERNED_NONCANONICAL` immediately?
15. Should the next approved task be a bounded component-integration plan with no empirical execution?

An acceptable approval must identify this report/version and occur after PDF creation. A reply such as "continue," earlier approval, or approval of the synthetic canary does not satisfy this report gate. Even explicit PDF approval does not authorize data acquisition, analysis, simulation, backtesting, or a research run.

Appendix A - Milestone timeline and decision status

Milestone	Result	Evidence class
Architecture audit/refinement	Modular target approved	Architecture
Slice A	Reproducible momentum infrastructure; non-actionable	Research-process
Slice A.5	Local equity dataset NOT TRUSTED	Data quality
Slice A.6	Provider-neutral ingestion ready; local dry run rejected	Infrastructure/data quality
Slices A.7-A.8	Public samples incomplete; population reconstruction failed	Source/sample qualification
Slices A.9-A.10	Official-response gate remains closed	Access governance
Slices A.11-A.12	Read-only current Kite coverage ready	Current-market infrastructure
R.1-R.2	Legacy evidence preserved and quarantined	Legacy evidence
R.3-R.4	Governed execution and one-use approval contracts	Governance infrastructure
R.5	Synthetic canary proposal	Infrastructure proposal
R.6	Canary executed once and independently audited PASS	Infrastructure evidence
R.7	Status PDF generated; owner review pending	Governance/documentation
R.7 amendment	Exact version2.0 commit statically reconciled; no merge or execution	Cross-repository governance

Important current decisions:

- Dataset: NOT TRUSTED for cross-sectional equity-edge research.
- Historical population: reconstruction incomplete.
- Official response: AWAITING_SUBSTANTIVE_OFFICIAL_RESPONSE.
- Canary: GOVERNED_CANARY_AUDIT_PASSED.
- Research report gate: REPORT_GENERATED_PENDING_REVIEW.
- Market research, acquisition, simulation, backtesting, and scoring: BLOCKED.

Appendix B - Cross-repository reconciliation

Field	custom_terminal	version2.0
URL	github.com/parikshithrg/custom_terminal.git	github.com/parikshithrg/version2.0.git
Branch and commit	main / 1a082bdad1667410c95931c4482971c133de918a	master / f9a6eae2cab1dd9e85d284e48b9863cae0b1298
Primary purpose	Governance, trust, evidence and canonical research control	Product dashboard, current displays and exploratory tools
Execution model	Exact governed gateway and one-use approval	Direct UI/CLI scripts and refresh controls
Data posture	Provider-neutral, immutable evidence and fail-closed capabilities	Current NIFTY 500, CSV/Yahoo/Kite and local F&O sources not historically qualified
Strategy results	Legacy evidence quarantined; no active edge	No meaningful benchmark-relative edge; results noncanonical
Retain	All authority and governance contracts	Selected UI, read-only display and provider patterns
Retire/isolate	None identified in this review	Persistent Kite secrets/tokens, live broker and direct action controls

Duplicates include current-data providers, Streamlit views, backtest abstractions, reports and data-refresh logic. Conflicts include security scope, credential persistence, recommendation semantics, validation labels and direct execution outside governance. The integration boundary is a published read model: version2.0-style views may consume approved custom_terminal artifacts but may not calculate or promote canonical evidence.

Unresolved matters are the eight repository-specific owner questions, data-use rights for the F&O and other local sources, which UI components to prioritize, and whether the live-action code should be removed or isolated. The complete machine-readable inventory is specs/cross_repository_reconciliation_v1.json.

Appendix C - Evidence and report index

Key sources used to reconcile this report:

- ARCHITECTURE_AUDIT.md and ARCHITECTURE_REFINEMENT.md.
- reports/SLICE_A_REPORT.md and Slice A.5 through A.12 reports.
- reports/DATASET_TRUST_REPORT.md and reports/HISTORICAL_DATA_REQUIREMENTS.md.
- reports/HISTORICAL_POPULATION_QUALIFICATION.md.
- reports/OFFICIAL_LIFECYCLE_QUALIFICATION.md.
- reports/BENCHMARK_AUDIT.md and benchmark public-data qualification.
- reports/COST_MODEL_AUDIT.md and historical cost qualification.
- reports/RESEARCH_SYSTEM_RECONCILIATION.md and R.2 through R.6 reports.
- reports/FUTURE_RUN_GOVERNANCE.md and reports/GOVERNED_EXECUTION_OPERATIONS.md.
- evidence/governance/canary_execution_anchor_v1.json.
- specs/pre_research_review_policy_v1.json.
- specs/cross_repository_reconciliation_v1.json.

Key hashes:

Object	SHA-256
Canary approval file	326afafdee722925c3fafdbbe3dc6b33ee61e0f2b7aaa48f564e8b6e993378e5
Canary root manifest	bdb3ee240ce2939dff6c523c305d5affdf9cd6b4deaf1fd0995efe675a9c40fb
Canary governance terminal event	9ddfaedf8fb0dd113417645b127c7dd3c021c478ae89b1648cf5e5d3caa5a0bb
Canary canonical record	63f9fa81774313355033841960a34c2f1ef6ef5c1ffdc69c34c0a652e6fa0c6e
Frozen legacy log	124886d439a90071a9b0f884851afbffd5dd783e52c2cedaeb4b2ac82eae1d
Neutral legacy ledger	79df7b785fef5025f605e0cef4a6dd49a039b034d66a92ea0aaafd99056fb392
Research-state fingerprint	40a18b14949d13b383dd549fe1d37c86881a61acf2d98bac9db2e6770d820e01

Appendix D - Glossary

Term	Plain-language meaning
Point-in-time	Uses only information that was knowable at the historical decision time.
Survivorship bias	Results look better because securities that later failed or disappeared are absent.
Stable identity	An internal security/listing ID that remains meaningful when symbols or names change.
Corporate action	Split, bonus, dividend, rights, merger, demerger, or identity change affecting price or ownership.
Terminal outcome	Economic treatment when a listing ends, merges, delists, or cannot be traded.
PRI	Price return index; ordinarily excludes reinvested cash dividends.
TRI	Total return index; includes defined dividend reinvestment.
Preregistration	A locked statement of the question, data, method, metric, and decision rule before outcome access.
Walk-forward	Repeated chronological training and later evaluation without using future information.
Canonical evidence	Evidence admitted through the governed manifest, approval, event, and import contracts.
Nonpromotable	Permanently barred from becoming production or actionable evidence.
Research-state fingerprint	One hash over every declared research-relevant file, used to detect staleness.
Report gate	Owner review prerequisite; it does not authorize a research run.

Known limitations and staleness policy

The fingerprint includes research and simulation code, contracts, specifications, data-trust declarations, proposals, governance policies, tracked evidence anchors, architecture decisions, and milestone reports that determine scientific scope. It excludes the generated PDF, its presentation source, the review record, mechanical PDF inventories, the R.7 completion report, tests, caches, and README. These exclusions prevent self-reference and allow documentation-only commits without automatic staleness.

Any substantive change to included code, contracts, data declarations, assumptions, universe, outcomes, costs, benchmarks, governance policy, proposal definition, or scientific-scope report changes the fingerprint and makes this PDF stale. A new PDF and post-generation owner review are then required.

Final owner reminder

Current review status: PENDING USER REVIEW. Research remains blocked. This document is ready to be read and corrected; it is not permission to analyze markets.

PRE_RESEARCH_PDF_READY_FOR_USER_REVIEW